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Editorial Office, *RSC Advances*
Royal Society of Chemistry
Thomas Graham House, Science Park
Milton Road, Cambridge CB4 0WF, UK

Dear Editor,

We submit our manuscript “**Experimental Inter-Report Scatter Bounds Machine-Learning Accuracy for TADF Singlet–Triplet Gaps—Structure and Semi-Empirical Features Reach the Same Limit**” for consideration as a Full Paper in *RSC Advances*.

What the paper reports. The singlet–triplet gap ΔE_{ST} is the photophysical observable that gates thermally activated delayed fluorescence (TADF). We measure where the limit on predicting that observable from cheap information lies—and find it is set by the experimental data, not by the model. On one controlled set of 231 donor–acceptor emitters (212 Bemis–Murcko scaffolds) we compare three levels of description against experiment: the 2D molecular structure, a semi-empirical excited-state calculation (GFN2-xTB/sTDA natural transition orbitals), and a range-separated TD-DFT reference (wB97X-D4/def2-SVP, Tamm–Dancoff, 231/231 converged, 609 core-hours). Structure alone reaches mean absolute error 0.091–0.100 eV; the semi-empirical excited state gives 0.096 eV; the TD-DFT gap gives 0.099 eV after in-fold calibration. They converge because the binding term is the label itself: independent measurements of the same compound scatter by 0.072 eV RMS, so the per-molecule median the model is trained on is known only to 0.049 eV—about half the model error. (Computed references carry a separate functional dependence of up to 0.58 eV between B3LYP and CAM-B3LYP; that bounds what a computed gap can mean and is not added into the experimental bound.) A further, unexpected result separates accuracy from ranking: under a source-paper (DOI-grouped) split, the semi-empirical features retain rank correlation the fingerprints lose ($\Delta\rho = 0.19$, 95 % CI 0.04 to 0.34, excluding zero narrowly and on a single split), so they order candidates across laboratories even where they do not improve the predicted value. This is a photochemistry benchmark with a physical-organic conclusion—what structural information does and does not encode about the exchange splitting—not solely a machine-learning exercise.

Contributions. (i) A controlled measurement of feature–target leakage for a derived-energy target. Its severity is known to depend on the model class; we quantify that here. Regressing ΔE_{ST} on features containing the S_1 and T_1 energies is circular ($\Delta E_{ST} \equiv E_{S_1} - E_{T_1}$): under cross-validation a linear model given those energies recovers the target exactly ($R^2 = 1.000$) against $R^2 = 0.36$ without them, while an axis-aligned random forest on the identical features reports only $R^2 = 0.59$ and clears cross-validation unremarked. That is how a tree-based screening study can carry this defect without the score looking wrong. (ii) Structure alone predicts the *experimental* ΔE_{ST} as accurately as the semi-empirical excited state—a scaffold-validated benchmark with a label-permutation null ($p = 0.001$), the two representations equal to within 0.017 eV on paired

folds. A new finding is the dissociation between accuracy and ranking under laboratory transfer: the semi-empirical features retain rank correlation under a source-paper split where fingerprints do not ($\Delta\rho = 0.19$, 95 % CI excluding zero narrowly, on a single split). (iii) A measured bound on the label itself—the median gap is known only to 0.049 eV, about half the model error—against which a 609-core-hour TD-DFT reference and a frozen pretrained molecular-transformer embedding both fail to beat the structure-only model. Applying a noise-ceiling calculation (Crusius *et al.*) to that precision gives $R_{\text{max}}^2 \approx 0.71\text{--}0.91$ depending on which noise floor is used; the observed $R^2 = 0.26$ reaches only 28–36% of that ceiling, so label imprecision bounds what is achievable but does not by itself explain the shortfall. (iv) A triage filter that enriches low-gap candidates by $1.2\text{--}1.4\times$ over random on held-out labelled data, though a bootstrap confidence interval shows this is resolved from chance only at the broader top-50/top-20 % cutoffs, not at the top-10/top-20 scale a laboratory would actually shortlist from—useful for coarse prioritisation, not a device-level or emission-spectrum predictor.

Why this is new. The advance is conceptual, not a better error. That ΔE_{ST} is reported inconsistently has been shown for individual emitters (Shuaib *et al.*, *J. Mater. Chem. C* 2026, **14**, 4353); we turn it into a quantitative inter-laboratory variability budget over 231 molecules and 173 source papers (single-report scatter 0.072 eV RMS, median-label precision 0.049 eV) and show it is the term that caps every predictor we test, semi-empirical or TD-DFT. Two further findings we have not seen reported: under a source-paper split a model’s accuracy advantage vanishes while the excited-state descriptors keep the rank order the fingerprints lose ($\Delta\rho = 0.19$, 95 % CI excluding zero narrowly, on a single split); and the feature–target leakage that inflates computed-gap models is fully exposed by a linear learner and substantially masked by a random forest. The multi-scale ML screening pipelines and the pretrained or multimodal TADF models of 2025–2026 make neither point, because none runs an experimental source-paper split or a controlled leakage comparison. The work is the prediction-side, statistical complement to the experimental measurement-methodology literature.

Prior consideration. An earlier version of this manuscript was reviewed at *Digital Discovery* (DD-ART-07-2026-000486) and declined on 27 August 2026. The three referee reports were constructive and we have used them to strengthen the paper. The decline cited insufficient novelty against existing TADF screening work; the paragraph above states what is new and what is not. An itemised response follows this letter. The main additions since that version are: the 609-core-hour range-separated TD-DFT reference run (Referee 1); a full data and code deposit on Zenodo, previously incomplete (Referee 2); a source-paper split, the removal of the library shortlist, and a trimmed inverted-gap discussion (Referee 3).

Open science. The experimental corpus, descriptor tables, trained models, SHAP analysis, TD-DFT outputs, and all Python code are openly available on Zenodo (DOI: 10.5281/zenodo.17436069) under a CC-BY-4.0 licence, with a SHA-256 manifest.

APC waiver. The corresponding author is affiliated with the University of Yaoundé 1, Cameroon, a Research4Life country; we are eligible for a full APC waiver under the RSC’s Research4Life arrangement and will provide documentation on request.

Suggested reviewers. None is a co-author or collaborator:

- Prof. Heather J. Kulik, Massachusetts Institute of Technology, USA (hjkulik@mit.edu) — ML for molecular design
- Prof. Andrew P. Monkman, Durham University, UK (a.p.monkman@durham.ac.uk) — TADF photophysics

- Prof. Olexandr Isayev, Carnegie Mellon University, USA (olexandr@cmu.edu) — ML for chemistry
- Prof. Denis Jacquemin, Nantes Université / CNRS CEISAM, France (Denis.Jacquemin@univ-nantes.fr) — TD-DFT excited-state benchmarks

The manuscript is not published elsewhere and is not under consideration by any other journal.
All authors have approved submission to *RSC Advances*.

Yours sincerely,

Jean-Pierre Tchapel Njafa
(on behalf of all authors)

Response to the reviewers

Manuscript: “Experimental Inter-Report Scatter Bounds Machine-Learning Accuracy for TADF Singlet–Triplet Gaps—Structure and Semi-Empirical Features Reach the Same Limit”

Prior review: *Digital Discovery* DD-ART-07-2026-000486

Referee comments are in italics. Every number cited below is reproducible from the Zenodo deposit (DOI: 10.5281/zenodo.17436069). Section and table numbers refer to the revised manuscript and its Supporting Information.

Referee 1

1.1 “*The dataset is tiny (<300 molecules), so it is not surprising that it doesn’t work.*”

Response. Dataset size is the object of study here, not an oversight. A correlation-test power analysis (Methods) shows that the observed Spearman $\rho = 0.36$ is resolved at 80% power with $n \approx 60$; at $n = 231$ the achieved power is ≈ 1 . We should be clear about the scope of that argument: it concerns the rank correlation only, and does not speak to the width of the R^2 interval, which we address separately and do not explain away. More directly, we now report a threefold corpus expansion to 728 experimental emitters through the same pipeline: the MAE *rises* from 0.098 to 0.124 eV, R^2 falls from 0.18 to -0.01 , and the rank correlation roughly halves (ρ 0.34 to 0.16) (§ Ceilings; SI Table, expansion). Adding data makes the prediction worse, which points at the label rather than the sample count—though we note in the manuscript that the added molecules come through a different structure-resolution pipeline, so chemical breadth and label quality are confounded in that particular test. The cleaner evidence is direct: the model reaches 0.074 eV on molecules whose replicate reports agree and 0.106 eV on those whose reports do not.

1.2 “*The correlation between experiment and theory (DFT or semi-empirical) is rather good ($r > 0.95$). It is not difficult to predict a computed singlet–triplet gap with ML and a large computed dataset.*”

Response. We now test this claim directly. A range-separated TD-DFT reference gap (wB97X-D4/def2-SVP, RIJCOSX, Tamm–Dancoff; ORCA 6; nroots 8, triplets included; GFN2-xTB geometry) was computed for all 231 molecules (609 core-hours, 16.1 min median per molecule; new Methods paragraph, new SI subsection S4.3). Against *experiment* the raw TD-DFT gap has Pearson $r = 0.43$ and $R^2 = -16.3$: it is the worst predictor in the feature-comparison table (MAE 0.654 eV), because it carries a systematic +0.65 eV offset. The $r > 0.95$ figures in the literature are theory-against-theory, or ML against a computed reference—the circular set-up diagnosed in contribution (i). Predicting the *measured* gap is the harder problem, and neither a semi-empirical nor a range-separated excited-state calculation solves it here.

1.3 “*Particularly strange is the refusal to use the computed gap as a descriptor while other computed quantities of similar cost are used. The semi-empirical computed gap is probably much better than the ML gap from structural parameters alone.*”

Response. We now use the computed gap as a feature, at both levels of theory, and report the result rather than omitting it (§ Benchmark; SI S4.3). The semi-empirical (sTDA) gap as an added feature does not change the error (NTO model 0.096 to 0.097 eV; Morgan model unchanged).

The TD-DFT gap as an added feature lowers the Morgan model from 0.091 to 0.087 eV—a paired 95% CI of 0.0004 to 0.0071 eV and Wilcoxon $p = 0.016$, so a real effect, but a 4% reduction bought with 609 core-hours. Used alone, the semi-empirical gap is *worse* than the structure-only forest (calibrated MAE 0.104 eV vs. 0.091 eV), contrary to the referee’s expectation. We have also replaced the earlier assertion that the feature sets are “statistically indistinguishable” with a measured equivalence bound: paired on identical folds, Morgan and NTO differ by 0.004 eV with a 95% CI of $[-0.017, +0.009]$ eV, so they are equivalent to within 0.017 eV rather than merely not separable. The energy scalars are excluded from the NTO descriptor set for a specific reason—including E_{S_1} and E_{T_1} as features is the circularity of contribution (i); the spatial NTO overlap descriptors we do keep encode exchange-integral physics without reducing to $E_{S_1} - E_{T_1}$.

1.4 “*I cannot see a practical use of this approach, and the didactic value is limited.*”

Response. The practical use is triage: on held-out labelled data the model enriches low-gap emitters by $1.2\text{--}1.4\times$ over random selection—precision 0.60 among the top 10 ranked candidates and 0.68 among the top 50, against a base rate of 0.49. (These are counts of molecules, not percentiles; the earlier submission conflated the two conventions and we have corrected it.) We have re-scoped every claim to that regime and removed the projection onto the unlabelled filtered library (see 3.4).

Two further results are directly usable. The first is a concrete recommendation: under a source-paper split the semi-empirical NTO descriptors hold Spearman $\rho = 0.35$ while fingerprints fall to 0.16, so a group that must rank candidates against measurements made in another laboratory should pay for the excited-state step, while a group ranking within its own data need not. That is a decision the cost of those descriptors can be weighed against.

The second is the didactic point, which we think is transferable. Under cross-validation a linear model given the two constituent energies recovers the target exactly ($R^2 = 1.000$) against $R^2 = 0.36$ without them. More usefully, how severe the leak *looks* depends on the learner: a random forest is axis-aligned, does not recover a difference of two continuous inputs exactly, and reaches only $R^2 = 0.59$ on identical leaked features. A tree-based study can therefore carry this defect while its cross-validated score appears unremarkable, which may be why the failure mode persists. Alongside it sits the calibrated chemistry ceiling: 2D structure and the semi-empirical excited state land within 0.017 eV of one another against experiment on paired folds, and a calibrated TD-DFT gap within a further 0.008 eV, while the raw uncalibrated TD-DFT gap overestimates by 0.65 eV.

Referee 2

2.1 “*Please provide a link to the Zenodo repository, and make sure the actual data is there. The .zip I downloaded only contained a README.md; the data was on GitHub. Please add it to Zenodo.*”

Response. The Zenodo record 10.5281/zenodo.17436069 now holds the full deposit, not only the README: the curated experimental corpus, the three descriptor tables, the trained random-forest models, the per-molecule SHAP output, the new TD-DFT reference outputs (tddft_reference_231.csv and timing), and every analysis and figure-generation script, with a manifest.csv giving a SHA-256 checksum for each file. The GitHub repository is retained as a mirror. The DOI is cited in the abstract, Methods, Conclusions, and the data-availability

statement.

Referee 3

3.1 *“Predictive capacity is extremely weak and statistically ambiguous: $R^2 = 0.25\text{--}0.31$ with 95% CIs spanning into negative territory. A model with zero-crossing R^2 intervals cannot be distinguished from a mean-value predictor. Spearman ρ is insufficient to claim predictive power.”*

Response. We agree and have made this the framing rather than a caveat. The abstract now states that the signal “sits just above a mean-value baseline (MAE 0.107 eV)”; the Conclusions say the advantage is not resolved under the harder split. What the signal *is*: it clears a label-permutation null ($p = 0.001$); the Spearman $\rho = 0.36$ has a 95% CI of 0.24 to 0.47 that excludes zero; the paired MAE advantage over the mean-value baseline excludes zero under scaffold CV. Actionable utility is therefore triage only ($1.2\text{--}1.4\times$ enrichment on labelled data, though a bootstrap confidence interval shows this is resolved from chance only at the top-50/top-20 % cutoffs and not at the top-10/top-20 scale a laboratory would actually shortlist from), and we no longer make any molecule-level selection claim. Under the source-paper (DOI-grouped) split introduced for this revision the advantage over the baseline is 0.005 eV with a CI that crosses zero—stated in the abstract, in § Benchmark, and as Limitations bullet (6). One result cuts the other way and is now reported: under that same split the semi-empirical NTO model holds Spearman $\rho = 0.35$ while the fingerprint model falls to 0.16, a paired difference of 0.19 with a 95% CI of [0.04, 0.34] that excludes zero. Ordering survives the change of protocol even where the predicted value does not, which is what a triage filter is asked to do.

We have also quantified, directly, how much of the residual gap even a perfect model could close. Applying the noise-ceiling framework of Crusius *et al.* to the label’s own measurement precision gives $R^2_{\text{max}} \approx 0.71\text{--}0.91$ (depending on whether the single-report or median-precision floor is used), so the observed $R^2 = 0.26$ reaches only 28–36% of what label imprecision alone would permit. Label noise therefore sets a real, quantified ceiling on R^2 , but it does not by itself explain why the model sits well below even that ceiling; we now state both halves of that argument explicitly (§ Ceilings).

3.2 *“The dataset aggregates literature values across unrecorded conditions (hosts, solvents, temperatures, techniques). Compiling medians from uncontrolled environments degrades the target. Test or discuss a solvent- and condition-controlled subset rather than blaming data noise.”*

Response. Methods now states explicitly that the corpus does not record measurement medium per value, so the per-molecule median is a consensus over uncontrolled conditions. We address the referee’s test with the repeat-measured subset: for the 102 molecules with independent reports, 58% have zero inter-report spread, and this cleaner-label subset gives the study’s *lowest* MAE, 0.080 eV (§ Benchmark). We must be careful with that number rather than offer it as support: a mean-value predictor restricted to the same 102 molecules scores 0.089 eV, so the model’s advantage there is 0.009 eV with a confidence interval spanning zero, and the low absolute error reflects the narrower spread of those labels rather than better prediction. We tested the referee’s proposition directly and it did not survive. An earlier “clean-label” subset of 208 molecules was mis-specified—it assigned zero spread to the 129 compounds with no replicate at all, discarding only 23. Restricting properly to replicated molecules gives 0.074 eV where reports agree and 0.106 eV where they do not, but a constant predictor fitted within each stratum

reproduces that ordering to within 0.001 eV, because the strata differ in the dispersion of the target. We therefore report no stratified evidence that label noise sets the ceiling, and say so in the manuscript. On the referee’s specific request we have gone further than stating a limitation. A strictly solvent-controlled subset cannot be built: phase and atmosphere are populated for 0 of the 1520 extracted rows. We therefore ran the closest test the data permits, splitting on source paper, since values drawn from a single publication share one laboratory’s conditions. The model’s advantage over a constant predictor fitted within each group is 0.0071 eV for the 218 single-source molecules and 0.0024 eV for the 13 multi-source ones, a difference of 0.005 eV whose 95% interval includes zero. The result is null and we report it as such, noting that 13 molecules give the test little power. We have also, throughout, compared every subset against a constant predictor fitted within that subset, having found that an earlier subset argument of ours was reproduced entirely by target dispersion.

3.3 *“With 212 scaffolds for 231 molecules, 92.9% of scaffolds are singletons, so the GroupK-Fold scaffold split is essentially a random split. On expansion to 728 molecules R^2 collapses to -0.01 and MAE worsens to 0.124 eV, showing the model does not generalise. The triage claim must be re-evaluated.”*

Response. The singleton fraction is now stated in Methods and the SI, and we have added a source-paper (DOI-grouped) GroupKFold over 173 papers as the harder transfer test: it holds out a synthetic series and a laboratory’s shared practice together. Under it the Morgan model gives MAE 0.102 eV ($R^2 = 0.18$) and its advantage over the mean-value baseline is not statistically resolved (§ Benchmark). The 728-molecule expansion is promoted to its own Results paragraph and read as the referee reads it—evidence against generalisation and for a label ceiling. The triage claim is now explicitly restricted to held-out labelled molecules, with the projection onto the filtered library removed.

3.4 *“The top-100 shortlist from the 22 194-molecule library has a split-conformal 90% interval of ± 0.18 eV, while the predicted gaps are 0.02–0.05 eV. The uncertainty engulfs the prediction; this offers no prioritisation value.”*

Response. We agree and have removed the shortlist. No ranked candidate list is deposited. The text (§ Library) now states that the split-conformal 90% interval spans 0.36 eV—about 7.6 times the predicted gap, a mean half-width of 0.18 eV—and that for every one of the 100 lowest-predicted molecules the interval’s lower bound is a negative (inverted) gap. Conformal prediction refuses molecule-level selection at this accuracy; the filtered set is kept only to indicate the scale of the chemical space the labelled-data triage would operate over. The corresponding SI table has been deleted.

3.5 *“NTO descriptors are computed at ground-state vertical geometries and miss excited-state relaxation and spin–orbit coupling. Dismissing the 0.195 eV adiabatic deviation as a rank-preserving offset is inadequate; this likely explains why NTO features do not beat 2D fingerprints.”*

Response. We have removed the dismissive wording. The 14-molecule vertical-vs-adiabatic comparison (CAM-B3LYP/def2-TZVP, MAE 0.195 eV) is now reported as a bound on the vertical approximation, not waved away, and the equivalence of NTO and 2D features is now stated as a measured equivalence bound (0.017 eV on paired folds) rather than as “mechanistically informative”. Limitations bullet (6) states that the descriptors are ground-state vertical, carry no

excited-state relaxation and no spin–orbit coupling, and that this is a plausible part of why they do not outperform 2D fingerprints. We agree that missing excited-state relaxation is a plausible contributor to the NTO ceiling and say so.

3.6 *“The single-determinant GFN2-xTB/sTDA framework enforces Aufbau occupation and cannot capture the double-excitation character of gap inversion; it predicts positive gaps for all inverted-gap candidates. The INVEST narrative should be streamlined or removed.”*

Response. Agreed, and the discussion is now short and factually corrected. A literature screen flags five corpus compounds with an inverted report; for two of them (HAP-3MF, TPA-NZP) the consensus median we train on is itself negative, and these are the only two negative labels among the 231. The model returns a positive gap for all five, so it carries the wrong sign for both molecules whose own label is negative. We now state that this is representational rather than statistical—the descriptors carry no signal that separates an inverted-gap molecule from a conventional one, since the semi-empirical step is computed under Aufbau ordering—and the “expected falsification” narrative is removed. The earlier submission described these as “external test cases”, which was wrong: they are inside the corpus.

3.7 *“An R^2 interval encompassing zero—does the model provide actionable predictive utility for materials discovery?”*

Response. For point prediction of ΔE_{ST} : no, and the manuscript now says so plainly—a noise-ceiling calculation (Crusius *et al.*) puts the maximum achievable R^2 at 0.71–0.91 given the label’s own measurement precision, and the observed $R^2 = 0.26$ reaches only 28–36% of that. For triage: yes, within a stated bound— $1.2\text{--}1.4\times$ enrichment of low-gap emitters over random selection on held-out labelled data, though a bootstrap CI resolves this from chance only at the top-50/top-20 % cutoffs, not at the top-10/top-20 scale, with conformal intervals that rule out single-molecule selection. The value of the paper is the calibrated ceiling and the leakage diagnosis, both of which are quantitative and reproducible.